

$$\text{Let } A = \begin{bmatrix} 6 & -2 \\ -3 & 1 \end{bmatrix}$$

$$a.) A - \lambda I = \begin{bmatrix} 6-\lambda & -2 \\ -3 & 1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (6-\lambda)(1-\lambda) - (-2)(-3) \\ = (6-7\lambda+\lambda^2) - 6 = \lambda^2 - 7\lambda$$

$$\lambda^2 - 7\lambda = 0 \quad \text{or} \quad \lambda(\lambda-7) = 0$$

$$b.) \lambda(\lambda-7) = 0 \rightarrow \lambda_1 = 0, \lambda_2 = 7 \quad (\lambda_1, \lambda_2) = (0, 7)$$

c.) Bases for eigenspaces:

$$1.) \lambda = 0, \text{ solve } Ax = 0: \begin{bmatrix} 6 & -2 & | & 0 \\ -3 & 1 & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & -1 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$3x - y = 0, \text{ Let } y = t, \text{ then } x = \frac{1}{3}t. \text{ Thus } x_1 = \begin{bmatrix} \frac{1}{3} \\ 1 \end{bmatrix} \approx \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$2.) \lambda = 7, \text{ Solve } (A - 7I)x = 0: A - 7I = \begin{bmatrix} -1 & -2 \\ -3 & -6 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -2 & | & 0 \\ -3 & -6 & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}, \quad x + 2y = 0 \rightarrow x = -2y$$

$$\text{Let } y = t, \text{ then } x = -2t$$

$$\text{Thus, } x_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$